

The Appendix for
**Fayaz M. Comparison of three functional regression methods on air pollution
throughout the first two COVID-19 lockdown phases across 31 Iranian provinces.
JAPH. 2026;11(2):217-230.**

<https://japh.tums.ac.ir/index.php/japh/article/view/1149>

Contents

Descriptive Statistics	3
Unadjusted.....	3
First Wave.....	3
Second Wave.....	8
Comparison between real value vs reconstructed.	13
Maps	15
CO.....	15
H ₂ O	16
SO ₂	18
AEI	19
HCHO.....	20

Descriptive Statistics

Unadjusted

First Wave

Table S1. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the first COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
01_AL	-0.0007; -2.50% (0.22;0.06)	54.3; 9.55% (0.42;0.1)	-0.000075; -37.44% (0.14;0.61)	0.0022; 1.46% (0.41;0.22)	-0.00003; -17.54% (0.72;0.62)	-0.31; 42.33% (0;0)	0.0000041;6.83% (0.68;0.53)
02_AR	-0.0018; -5.86% (0;0.01)	-44.4; -6.36% (0.43;0.52)	-0.000011;- 14.9% (0;0)	0.003; 1.92% (0.21;0.24)	0.000004; 5.26% (0.94;0.67)	-0.22; 24.29% (0.03;0.01)	-0.0000042;-7.65% (0.74;0.87)
03_AE	-0.0012; -4.14% (0;0)	52.8; 9.72% (0.19;0.31)	-0.000009;- 12.53% (0;0)	0.0021; 1.34% (0.38;0.43)	0.000213;191.89% (0.35;0.98)	-0.27; 32.54% (0;0)	0.0000067;19.2% (0.35;0.71)
04_AW	-0.0008; -2.84% (0.08;0.12)	74.4; 15.25% (0.05;0.06)	-0.000008;- 11.79% (0;0)	0.0015; 0.96% (0.54;0.66)	0.000135;195.65% (0.34;0.93)	-0.27; 33.53% (0;0)	0.0000126;46.67% (0.15;0.52)
05_BU	-0.0002; -0.60% (0.59;0.85)	-118.7;- 8.28% (0.34;0.77)	-0.000001;- 1.67% (0.64;0.39)	0.008; 6.28% (0;0)	0.000099;48.06% (0.17;0.28)	-0.15; 48.57% (0.21;0.1)	0.0000021;2.56% (0.81;0.97)
06_CB	-0.0005; -1.96% (0.31;0.5)	49.8; 9.89% (0.46;0.15)	-0.000004;- 6.75% (0.02;0.01)	0.0054; 3.96% (0.01;0.02)	0.000043; 74.14% (0.46;0.33)	-0.32; 68.65% (0;0)	-0.000003;-7.81% (0.67;0.6)
07_FA	-0.0004; -1.41%	-76.3; -7.99%	-0.000001;- 1.63%	0.0086; 6.81%	0.000002; 1.8%	-0.31; 64.86%	-0.0000063; -9.98%

Table S1. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the first COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
	(0.29;0.35)	(0.38;0.76)	(0.68;0.09)	(0;0)	(0.96;0.35)	(0;0)	(0.29;0.2)
08_GI	-0.0012; -3.69% (0.05;0.03)	0.8; 0.09% (0.99;0.77)	-0.000014; -14.27% (0.05;0.01)	0.0028; 1.8% (0.28;0.33)	0.00009; 128.57% (0.27;0.26)	-0.12; 14.34% (0.28;0.26)	-0.0000026; -4.19% (0.87;0.77)
09_GO	-0.0002; -0.61% (0.77;0.3)	19.9; 2.02% (0.83;0.63)	-0.000008; -10.99% (0;0)	0.0049; 3.26% (0.06;0)	-0.00003; -26.55% (0.63;0.43)	-0.21; 27.3% (0.03;0.02)	-0.0000174; -23.93% (0.08;0.07)
10_HA	-0.0007; -2.57% (0.12;0.17)	60.1; 11.27% (0.26;0.17)	-0.000012; -15.99% (0.01;0)	0.0018; 1.21% (0.48;0.47)	0.000054; 79.41% (0.53;0.82)	-0.39; 54.11% (0;0)	0.0000025;7.02% (0.76;0.62)
11_HO	-0.0001; -0.31% (0.77;0.98)	10.4; 0.7% (0.94;0.87)	-0.000004; -6.8% (0.01;0.01)	0.0096; 7.85% (0;0)	0.000031; 39.24% (0.4;0.38)	-0.22; 97.56% (0.05;0.05)	-0.0000021; -2.74% (0.71;0.68)
12_IL	-0.0006; -1.86% (0.16;0.23)	27; 2.62% (0.8;0.68)	-0.000007; -10.73% (0;0)	0.0019; 1.31% (0.44;0.46)	0.00024; 135.59% (0.23;0.31)	-0.28; 40.89% (0.02;0.01)	-0.0000051; -8.79% (0.58;0.93)
13_IS	-0.0005; -1.75% (0.35;0.4)	110.6; 16.04% (0.15;0.05)	-0.000005; -6.38% (0.01;0.02)	0.0064; 4.62% (0;0)	-0.000009; -9.68% (0.83;0.8)	-0.37; 68.71% (0;0)	0.0000009;1.52% (0.88;0.56)
14_KE	-0.0001; -0.35% (0.75;0.49)	51.4; 5.43% (0.58;0.56)	-0.000004; -5.65% (0.04;0.02)	0.0099; 7.84% (0;0)	0.00003; 34.09% (0.36;0.06)	-0.31; 64.07% (0;0)	-0.0000087; -12.61% (0.14;0.16)
15_KM	-0.0009; 	55; 	-0.000007; 	0.0008; 	0.000122; 	-0.3; 	0.0000093;24.35%

Table S1. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the first COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
	-3.04% (0.03;0.05)	7.41% (0.45;0.51)	-9.75% (0.01;0.01)	0.54% (0.75;0.96)	137.08% (0.31;0.68)	35.71% (0;0)	(0.3;0.2)
16_KN	-0.0008; -2.67% (0.05;0.05)	-50.5; -6.07% (0.51;0.91)	-0.000006; -9.28% (0.02;0)	0.0055; 3.74% (0.04;0)	-0.000068; -144.68% (0.13;0.13)	-0.34; 48.57% (0;0)	-0.0000032; -5.96% (0.72;0.31)
17_KR	-0.0006; -2.00% (0.08;0.17)	-32.3; -3.29% (0.72;0.95)	-0.000004; -5.71% (0.04;0.02)	0.0071; 5.02% (0;0)	0.000021; 39.62% (0.58;0.49)	-0.4; 59.73% (0;0)	-0.0000043; -7.58% (0.49;0.57)
18_KS	-0.0004; -1.38% (0.28;0.32)	48.8; 5.06% (0.56;0.47)	-0.000003; -4.21% (0.07;0.09)	0.01; 7.57% (0;0)	0.000004; 10% (0.9;0.04)	-0.41; 84.27% (0;0)	-0.0000073; -13.06% (0.32;0.73)
19_KH	-0.0003; -0.89% (0.47;0.57)	-82.2; -6.08% (0.59;0.99)	-0.000002; -3.12% (0.33;0.14)	0.0047; 3.41% (0.02;0.02)	0.00023; 138.55% (0.11;0.06)	-0.17; 29.29% (0.09;0.02)	-0.0000012; -1.28% (0.88;0.92)
20_KB	-0.0008; -2.86% (0.2;0.39)	71.5; 8.92% (0.51;0.43)	-0.000005; -7.79% (0;0)	0.0068; 5.17% (0;0)	0.000099; 76.15% (0.22;0.34)	-0.22; 40.79% (0.02;0.01)	-0.0000152; -27.89% (0.15;0.51)
21_KU	-0.001; -3.62% (0.01;0.01)	38.2; 7.46% (0.4;0.33)	-0.000007; -10.85% (0.01;0)	0.0018; 1.18% (0.48;0.7)	0.000134; 291.3% (0.23;0.13)	-0.36; 46.36% (0;0)	0.0000143;71.5% (0.07;0.14)
22_LO	-0.0007; -2.50% (0.18;0.15)	30.5; 4.42% (0.71;0.45)	-0.000006;- 9.02% (0.01;0)	0.0021; 1.45% (0.4;0.47)	0.000106; 100% (0.41;0.87)	-0.43; 64.83% (0;0)	0.000003;7.65% (0.71;0.46)

Table S1. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the first COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
23_MA	-0.0004; -1.47% (0.32;0.33)	63.9; 10.3% (0.38;0.11)	-0.000022;- 23.4% (0.01;0.01)	0.0022; 1.5% (0.38;0.41)	0.000018; 12.08% (0.8;0.72)	-0.43; 58.47% (0;0)	0.0000072;14.66% (0.32;0.32)
24_MD	-0.0014; -4.68% (0.01;0.02)	0.5; 0.07% (1;0.63)	-0.000014; -13.82% (0.03;0.01)	0.004; 2.65% (0.13;0.05)	-0.000044; -23.16% (0.59;0.16)	-0.22; 28.13% (0.01;0.01)	-0.0000096; -10.12% (0.52;0.93)
25_QA	-0.0009; -3.16% (0.09;0.11)	109.1; 18.65% (0.08;0.09)	-0.000018; -17.41% (0.03;0.14)	0.0021; 1.39% (0.41;0.3)	-0.000002; -1.55% (0.99;0.74)	-0.28; 31.37% (0.01;0.01)	-0.0000099; -22.65% (0.38;0.79)
26_QO	-0.0001; -0.33% (0.82;0.78)	75.7; 8.55% (0.45;0.05)	-0.000029; -25.42% (0.02;0)	0.003; 2.05% (0.23;0.18)	-0.000027; -13.37% (0.72;0.75)	-0.3; 42.53% (0.01;0.01)	-0.0000061; -7.26% (0.38;0.31)
27_SE	-0.0007; -2.24% (0.05;0.05)	69.1; 8.35% (0.34;0.09)	-0.000007; -9.9% (0;0)	0.0059; 4.09% (0.02;0)	-0.00002; -23.53% (0.62;0.73)	-0.34; 51.93% (0;0)	-0.0000062; -9.41% (0.28;0.28)
28_SB	0; 0.00% (0.92;0.99)	-26.7; -2.17% (0.8;0.92)	-0.000004; -6.56% (0;0)	0.0106; 8.62% (0;0)	0.000003; 8.82% (0.9;0.47)	-0.3; 108.02% (0.01;0.01)	-0.0000056; -7.97% (0.22;0.25)
29_TE	-0.0009; -3.09% (0.1;0.14)	62; 10.27% (0.34;0.07)	-0.000056; -29.48% (0.01;0)	0.003; 2.02% (0.25;0.11)	0.000094; 40.34% (0.6;0.63)	-0.27; 37.61% (0;0)	-0.0000023; -2.96% (0.81;0.43)
30_YA	-0.0003; -1.02%	105; 12.81%	-0.000003; -5.12%	0.0085; 6.35%	-0.000001; -1.33%	-0.41; 91.05%	-0.0000033; -5.39%

Table S1. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the first COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
	(0.36;0.32)	(0.16;0.12)	(0.05;0.01)	(0;0)	(0.98;0.35)	(0;0)	(0.46;0.5)
31_ZA	-0.0009; -3.23% (0.03;0.04)	74.2; 13.9% (0.16;0.11)	-0.000012; -16.26% (0.01;0)	0.0022; 1.44% (0.4;0.54)	0.00009; 225% (0.21;0.79)	-0.28; 32.84% (0;0)	0.0000033;8.11% (0.66;0.89)
Estimated Mean Differences [$\mu_{COVID}-\mu_{Previous}$], growth rate (%) (P-Value for T-test unequal Variance; P-Value for Wilcoxon)							

Second Wave

Table S2. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the second COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
01_AL	-0.0013; -4.41% (0.17;0.09)	-242.3; -27.32% (0.23;0.22)	0.000024; 17.44% (0.2;0.15)	-0.0142; -9.24% (0;0)	-0.0001; -102.15% (0.23;0.38)	0.18; -15.77% (0.24;0.14)	0.0000312; 39.39% (0.05;0.08)
02_AR	-0.0003; -0.99% (0.71;0.98)	185.1; 24.32% (0.19;0.35)	0.000005; 7.07% (0.03;0.03)	-0.0155; -9.81% (0;0)	-0.00003; -41.43% (0.46;0.57)	-0.23; 19.91% (0.13;0.2)	0.000017; 30.8% (0.08;0.05)
03_AE	-0.0008; -2.72% (0.18;0.22)	29; 3.92% (0.84;1)	0.000006; 8.96% (0.01;0.02)	-0.0152; -9.7% (0;0)	-0.00004; -42% (0.52;0.51)	-0.03; 2.35% (0.81;0.76)	0.0000145; 30.08% (0.04;0.06)
04_AW	-0.0009; -3.12% (0.11;0.27)	52.9; 7.35% (0.75;0.8)	0.000007; 9.79% (0;0)	-0.0144; -9.22% (0;0)	0.00003; 54.24% (0.59;0.48)	-0.09; 7.75% (0.41;0.44)	0.0000089; 21.71% (0.24;0.32)
05_BU	0.0003; 0.86% (0.59;0.4)	-439.1; -30.03% (0.01;0.02)	0.000013; 20.16% (0;0)	0.0001; 0.08% (0.93;0.63)	0.00028; 106.51% (0.01;0.01)	-0.06; 7.34% (0.79;0.84)	0.0000438; 54.48% (0;0)
06_CB	-0.0005; -1.86% (0.15;0.15)	27.7; 4.62% (0.77;0.93)	0.000007; 11.97% (0;0)	-0.0043; -3.07% (0.15;0.07)	0; -6.52% (0.95;0.76)	-0.32; 37.13% (0.11;0.22)	0.0000241; 57.79% (0.01;0.01)
07_FA	0; 0.00% (0.9;0.93)	-292.3; -26.89% (0.03;0.03)	0.00001; 15.87% (0;0)	0.0005; 0.38% (0.8;0.89)	0.00015; 133.03% (0;0)	-0.07; 7.13% (0.75;0.67)	0.000031; 50.65% (0;0)
08_GI	-0.0007; 	181.7; 	0.000012; 	-0.0161; 	0.00002; 	-0.22; 	0.0000248;

Table S2. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the second COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
	-2.08% (0.43;0.32)	15.53% (0.34;0.41)	14.34% (0.01;0.01)	-10.18% (0;0)	13.68% (0.86;0.88)	19.2% (0.33;0.29)	35.08% (0.2;0.42)
09_GO	-0.0009; -2.63% (0.21;0.2)	335.7; 33.82% (0.07;0.09)	0.000008; 11.62% (0;0)	-0.0126; -8.11% (0;0)	0.00002; 40.43% (0.67;0.74)	-0.07; 6.77% (0.71;0.84)	0.0000247; 44.67% (0.21;0.29)
10_HA	-0.0012; -4.24% (0.02;0.02)	-104; -13.54% (0.4;0.51)	0.000009;14.04% (0;0.01)	-0.013; -8.57% (0;0)	-0.00004; -61.54% (0.41;0.48)	0.14; -11.08% (0.44;0.2)	0.000025; 48.26% (0.01;0.02)
11_HO	-0.0004; -1.20% (0.46;0.67)	-933.4; -45.46% (0;0)	0.000007;11.79% (0;0)	0.0023; 1.8% (0.16;0.27)	0.00031; 364.29% (0;0)	0.22; -32.35% (0.24;0.38)	0.0000296; 44.05% (0;0)
12_IL	-0.0012; -3.58% (0.18;0.13)	-233.3; -21.66% (0.16;0.24)	0.000007;11.65% (0;0)	-0.0091; -6.17% (0;0.01)	0; -1.61% (0.97;1)	0.1; -10.46% (0.41;0.22)	0.0000049; 5.88% (0.64;0.74)
13_IS	-0.0012; -3.96% (0.02;0.02)	-291.7; -28.5% (0.05;0.07)	0.000009;13.49% (0;0)	-0.0062; -4.31% (0.04;0.03)	-0.00001; -15.19% (0.79;0.71)	0.08; -8.6% (0.58;0.44)	0.0000245; 46.93% (0;0.01)
14_KE	-0.0013; -4.28% (0.04;0.02)	-474.7; -35.51% (0;0)	0.000004;6.23% (0.08;0.3)	0.0032; 2.44% (0.09;0.14)	0.00004; 35.29% (0.34;0.51)	0.05; -5.44% (0.75;0.98)	0.0000208; 39.02% (0.04;0.03)
15_KM	-0.0017; -5.57% (0;0)	-163.4; -19.04% (0.2;0.29)	0.000006; 9.2% (0.01;0.02)	-0.0126; -8.33% (0;0)	-0.00003;-29.91% (0.6;0.84)	0.12; -9.02% (0.42;0.24)	0.0000081; 13.04% (0.38;0.33)

Table S2. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the second COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
16_KN	-0.0009; -2.92% (0.05;0.02)	147.5; 17.35% (0.25;0.24)	0.000007; 11.2% (0;0)	-0.0102; -6.71% (0;0.01)	-0.00001; -130% (0.52;0.76)	0.01; -1.02% (0.95;0.93)	0.0000197; 45.6% (0.05;0.09)
17_KR	-0.0011; -3.55% (0.01;0.02)	-153.7; -15.8% (0.12;0.22)	0.000007;11.56% (0;0)	-0.0038;-2.61% (0.16;0.14)	0.00003;57.78% (0.43;0.63)	0.18;- 16.88% (0.11;0.14)	0.0000174; 36.1% (0.05;0.09)
18_KS	-0.0015; -4.95% (0.01;0.01)	-434.3; -35.56% (0;0)	0.000002;3.25% (0.15;0.15)	0.0031;2.27% (0.17;0.16)	-0.00003;-50.91% (0.17;0.13)	0.18;- 17.92% (0.2;0.32)	0.0000126; 25.3% (0.13;0.14)
19_KH	-0.0005; -1.44% (0.38;0.14)	-436.3; -32.86% (0.04;0.13)	0.000018; 25.67% (0;0)	-0.004; -2.83% (0.16;0.09)	0.00008; 30.83% (0.27;0.17)	-0.14; 18.19% (0.33;0.55)	0.0000199; 18.43% (0.15;0.1)
20_KB	-0.0001; -0.34% (0.9;0.79)	-143.4; -14.62% (0.48;0.79)	0.000012; 20.21% (0;0)	-0.0014; -1.03% (0.61;0.41)	0.00002; 10.43% (0.84;0.35)	-0.3; 31.64% (0.16;0.24)	0.0000462; 83.54% (0.02;0)
21_KU	-0.0008; -2.85% (0.08;0.07)	-90.6; -13.22% (0.38;0.44)	0.000007; 11.51% (0;0)	-0.015; -9.72% (0;0)	0.00006;75% (0.55;0.98)	0.09;-7.52% (0.53;0.32)	0.0000222; 61.84% (0.02;0.04)
22_LO	-0.0017; -5.78% (0;0)	-187.3; -22.49% (0.09;0.09)	0.000011; 18.09% (0;0)	-0.0094; -6.4% (0;0.01)	0; -4.29% (0.96;0.93)	0.16;- 12.78% (0.29;0.11)	0.0000139; 25% (0.16;0.22)
23_MA	-0.0013; -4.55% (0;0.01)	-177.3; -21.36% (0.13;0.2)	0.000013; 18.23% (0;0)	-0.0115; -7.67% (0;0)	0; -2.25% (0.97;0.71)	0.29; -22.05% (0.06;0.08)	0.0000224; 41.79% (0;0)

Table S2. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the second COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
24_MD	-0.0011; -3.56% (0.17;0.22)	217.7; 25.2% (0.16;0.16)	0.000016; 18.86% (0;0)	-0.0139; -8.95% (0;0)	0.00002; 23.53% (0.78;0.38)	-0.09; 7.86% (0.62;0.59)	0.0000163; 18.36% (0.36;0.45)
25_QA	-0.0013; -4.38% (0.05;0.07)	-140.7; -15% (0.36;0.48)	0.000016; 20.15% (0.01;0)	-0.0141; -9.17% (0;0)	-0.00004; -51.35% (0.5;0.48)	0.16; -12.03% (0.31;0.2)	0.0000276; 52.47% (0.05;0.03)
26_QO	-0.0016; -4.94% (0;0)	-435.9; -36.59% (0.01;0)	0.000011; 12.88% (0.01;0.01)	-0.0109; -7.24% (0;0)	-0.00004; -44.33% (0.41;0.44)	0.53; -45.98% (0;0)	0.0000134; 16.63% (0.27;0.15)
27_SE	-0.0013; -4.06% (0.01;0.02)	-175.5; -16.56% (0.1;0.24)	0.000005; 7.38% (0.02;0.03)	-0.0094; -6.25% (0.01;0.01)	-0.00002; -52.17% (0.35;0.35)	0.44; -39.22% (0;0.01)	0.00001; 17.36% (0.32;0.24)
28_SB	-0.0009; -2.88% (0.2;0.07)	-732.3; -45.65% (0;0)	0.000003; 5.03% (0.03;0.04)	0.004; 3.1% (0.03;0.04)	0.00016; -2285.71% (0.09;0.22)	0.29; -39.17% (0.15;0.16)	0.0000084; 14.74% (0.16;0.13)
29_TE	-0.0018; -5.79% (0.01;0.05)	-212.2; -23.83% (0.11;0.07)	0.000018; 11.66% (0.23;0.24)	-0.0128; -8.38% (0;0)	0.00001; 15.48% (0.83;0.93)	0.25; -23.14% (0.11;0.09)	0.0000181; 23.51% (0.2;0.2)
30_YA	-0.0012; -3.90% (0.06;0.02)	-273.5; -25.02% (0.01;0)	0.000004; 6.85% (0.02;0.04)	-0.0009; -0.65% (0.76;0.63)	-0.00001; -16.44% (0.67;0.6)	0.1; -10.53% (0.57;0.67)	0.000012; 22.26% (0.15;0.2)
31_ZA	-0.0004; -1.41% (0.42;0.44)	-107.6; -13.81% (0.43;0.59)	0.000006; 9.6% (0;0)	-0.0151; -9.74% (0;0)	-0.00003; -55.77% (0.48;0.37)	0.08; -6.89% (0.59;0.24)	0.0000292; 63.89% (0.05;0.03)

Table S2. Estimated mean differences, growth rates (%), and corresponding P-values for air pollution indices across Iranian provinces during the second COVID-19 lockdown wave. Each cell presents the estimated mean difference [$\mu_{COVID} - \mu_{Previous}$] followed by the relative growth rate (in %) and the P-values from both unequal-variance t-tests and Wilcoxon tests (shown in parentheses).

	CO	H ₂ O	NO ₂	O ₃	SO ₂	AEI	HCHO
Estimated Mean Differences [$\mu_{COVID}-\mu_{Previous}$], growth rate (%) (P-Value for T-test unequal Variance; P-Value for Wilcoxon)							

Comparison between real value vs reconstructed.

Table S3. Quarterly averages of real and reconstructed values for air pollution and meteorological indices across two COVID-19 lockdown waves in Iran. The table presents observed (Real) and model-reconstructed (Reconstructed) values for each gas and climate variable (CO, CO₂, NO₂, O₃, SO₂, AEI, HCHO, Pressure, Total Precipitation, Mean Temperature, and U-Wind) across three quantile (Q1–Q3) within each wave, comparing “No COVID” and “COVID” conditions.

GAS	Waves	Status	Real			Reconstructed		
			Q1	Q2	Q3	Q1	Q2	Q3
CO	Wave 1	No COVID	29.06	29.62	30.35	29.52	29.59	29.67
		COVID	27.96	28.87	29.95	28.83	28.89	29.03
	Wave 2	No COVID	30.52	30.74	31.19	30.74	30.75	30.76
		COVID	29.08	29.33	30.45	29.69	29.76	29.86
CO ₂	Wave 1	No COVID	66.30	79.32	100.94	80.75	82.39	85.62
		COVID	73.39	82.04	93.52	83.34	84.45	85.38
	Wave 2	No COVID	81.11	113.42	118.78	100.14	102.03	104.46
		COVID	71.86	91.34	93.13	82.21	83.31	84.56
NO ₂	Wave 1	No COVID	70.98	75.04	83.60	80.07	81.53	82.15
		COVID	64.77	68.61	74.39	68.23	69.12	69.77
	Wave 2	No COVID	69.93	71.69	72.94	71.39	71.57	71.78
		COVID	77.24	80.61	82.50	80.54	80.93	81.27
O ₃	Wave 1	No COVID	13.56	14.37	14.93	14.19	14.31	14.37
		COVID	14.34	14.66	15.12	14.74	14.80	14.86
	Wave 2	No COVID	14.32	14.50	14.91	14.62	14.66	14.69
		COVID	13.23	13.91	14.20	13.78	13.85	13.91
SO ₂	Wave 1	No COVID	4.49	9.62	14.44	9.89	10.58	11.07
		COVID	6.93	12.35	19.03	-15.42	12.76	32.37
	Wave 2	No COVID	5.05	8.86	12.47	8.87	9.25	9.50
		COVID	8.67	10.93	13.38	11.31	11.51	11.75
AEI	Wave 1	No COVID	-7.57	-6.10	-5.07	-6.63	-6.55	-6.44
		COVID	-10.70	-9.87	-7.58	-9.51	-9.36	-9.23
	Wave 2	No COVID	-11.30	-10.31	-9.85	-10.70	-10.57	-10.50
		COVID	-11.53	-9.98	-8.89	-10.02	-9.92	-9.88
HCHO	Wave 1	No COVID	48.91	59.74	65.95	57.00	57.99	59.51
		COVID	45.44	53.94	63.75	52.37	53.67	55.29
	Wave 2	No COVID	51.02	62.83	70.34	58.87	60.11	60.59
		COVID	74.15	78.44	86.63	80.42	80.93	81.23
Pressure	Wave 1	No COVID	86.51	86.68	86.80	86.64	86.65	86.66
		COVID	86.64	86.84	87.04	86.80	86.81	86.84
	Wave 2	No COVID	86.75	86.85	86.90	86.84	86.85	86.85
		COVID	87.00	87.04	87.11	87.03	87.04	87.04
TPrec	Wave 1	No COVID	10.13	24.49	50.18	40.28	42.76	47.25

		COVID	3.20	9.90	40.75	26.79	29.47	30.39
	Wave 2	No COVID	1.67	3.06	14.13	9.47	11.25	12.14
		COVID	1.52	2.21	5.01	3.30	3.47	3.53
MTemp	Wave 1	No COVID	28.33	28.55	28.83	28.51	28.56	28.60
		COVID	28.41	28.50	28.65	28.50	28.51	28.54
	Wave 2	No COVID	28.72	28.89	29.03	28.82	28.84	28.86
		COVID	29.10	29.30	29.48	29.27	29.29	29.32
UWind	Wave 1	No COVID	2.65	3.21	3.60	3.18	3.21	3.30
		COVID	2.60	3.24	3.66	3.17	3.20	3.25
	Wave 2	No COVID	2.37	2.51	3.11	2.72	2.79	2.84
		COVID	2.63	2.98	3.10	2.95	2.98	3.00

Maps

CO

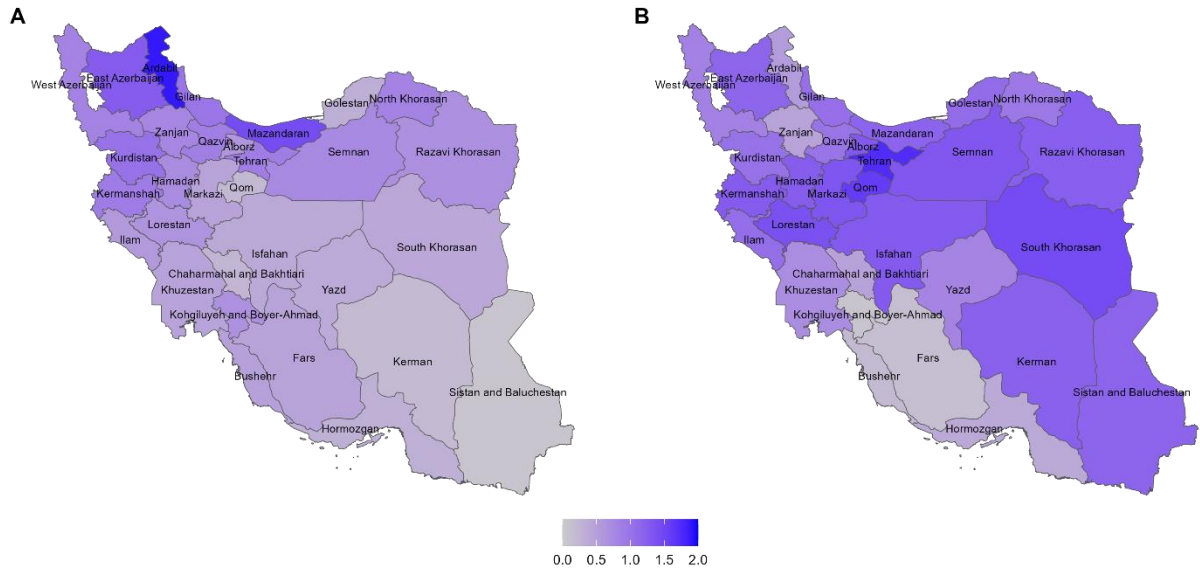


Figure S1. Differences between observed and predicted values obtained from the ffpc-based Functional-On-Functional regression (pffr()) model for CO (×1,000) during the COVID-19 lockdowns in Iran. Panel (A) represents the first wave, and Panel (B) represents the second wave.

H₂O

A



B

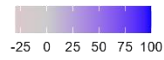


Figure S2. Differences between observed and predicted values obtained from the ffpc-based Functional-On-Functional regression (pffr()) model for H₂O ($\div 10$) during the COVID-19 lockdowns in Iran. Panel (A) represents the first wave, and Panel (B) represents the second wave.

O₃

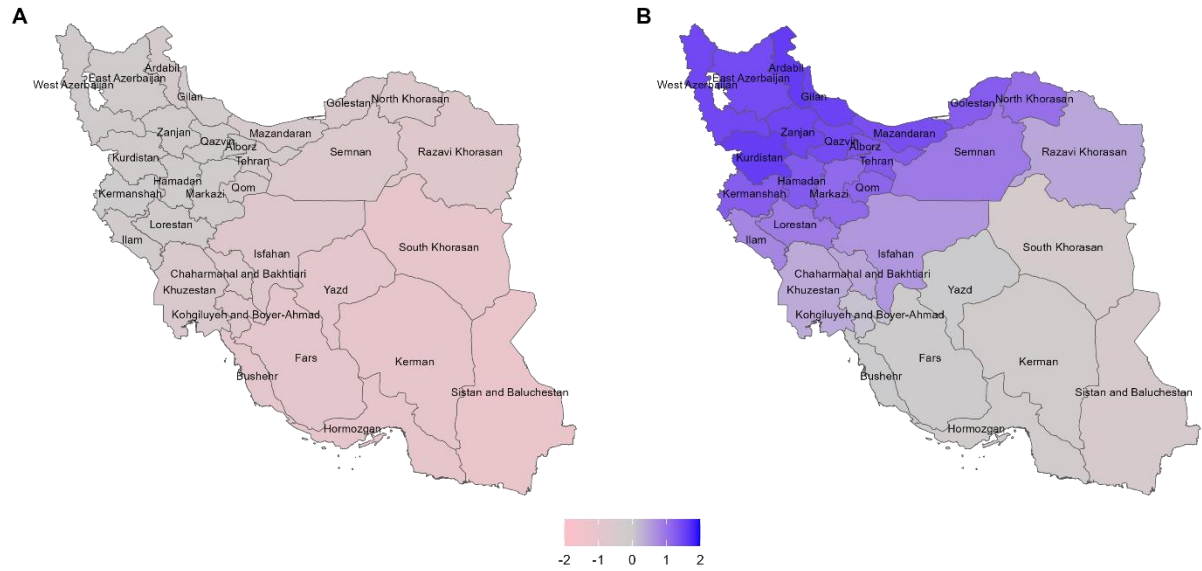
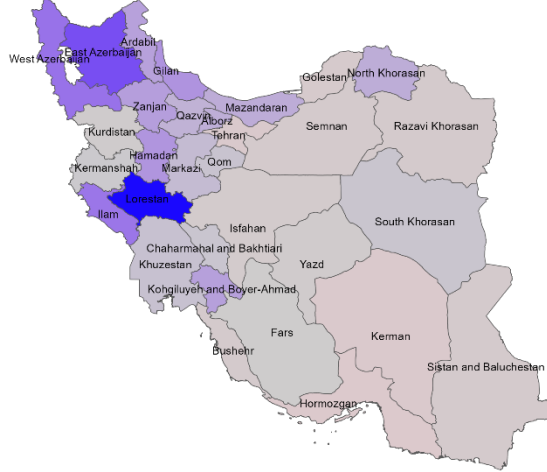


Figure S3. Differences between observed and predicted values obtained from the ffpc-based Functional-On-Functional regression (pfpr()) model for O₃ ($\times 100$) during the COVID-19 lockdowns in Iran. Panel (A) represents the first wave, and Panel (B) represents the second wave.

SO₂

A



B

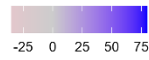


Figure S4. Differences between observed and predicted values obtained from the ffpc-based Functional-On-Functional regression (pffr()) model for SO₂ ($\times 100,000$) during the COVID-19 lockdowns in Iran. Panel (A) represents the first wave, and Panel (B) represents the second wave.

AEI

A



B



Figure S5. Differences between observed and predicted values obtained from the ffpc-based Functional-On-Functional regression (pffr()) model for AEI ($\times 10$) during the COVID-19 lockdowns in Iran. Panel (A) represents the first wave, and Panel (B) represents the second wave.

A



(A) represents the first wave, and Panel (B) represents the second wave.